BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

PROJECT

Project Title:

4 to 16 Decoder
Using Op-Amp

Course No. : EEE 208 S

Course Name: Electronic Circuits II Laboratory

Date Of Submission:
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Section: B1

Level 2 Term 2; Dept: EEE

Project Title: *4 to 16 Decoder Using Op-Amp*

Objectives :

- To implement logic gates using Op-Amp is the main objective of our project.
- To implement an application of logic gate – which is the decoder circuit in our project.

Software Used: PSpice 9.2

Components Used:

- Resistor
- VDC
- Op-Amp : uA741
- Gnd_earth

Theory:

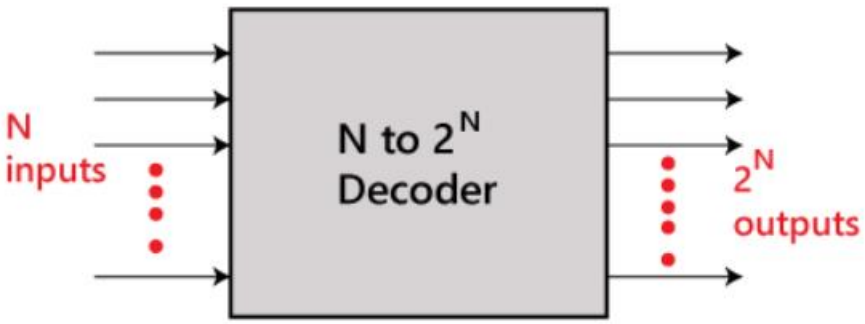


Fig : Block Diagram of a Decoder

The combinational circuit that change the binary information into 2^N output lines is known as **Decoders.** The binary information is passed in the form of N input lines. The output lines

define the 2^N-bit code for the binary information. At a time, only one input line is activated for simplicity. The produced 2^N-bit output code is equivalent to the binary information.

4 To 16 Decoder:

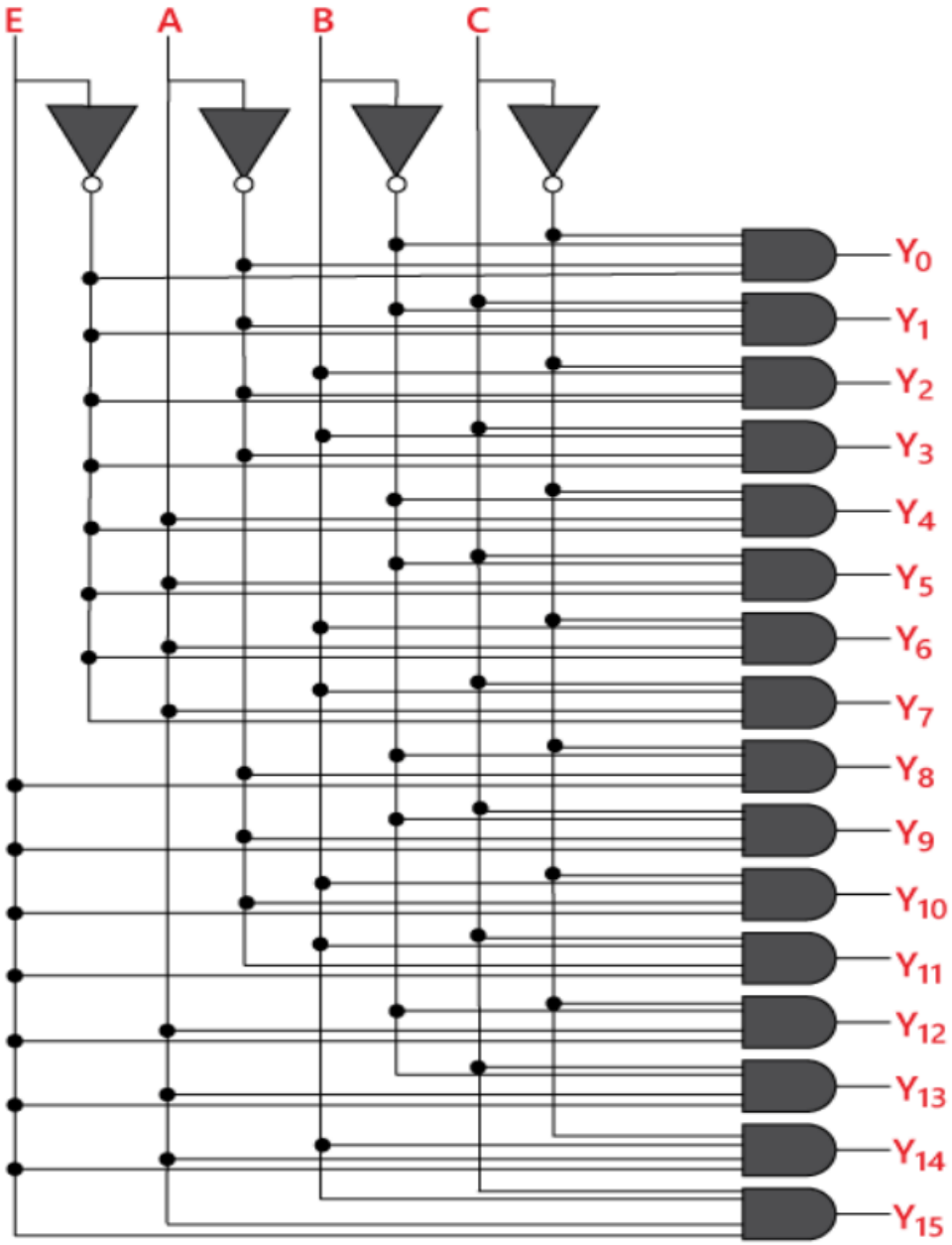
This decoder takes a 4-bit binary input and generates a 16-bit output, where only one of the 16 output lines is active (high) based on the binary value of the input. Each output line represents a unique combination of the input bits.

In the input stage, the 4-bit binary input is fed into the decoder circuit. Each bit is connected to an op-amp configured as a comparator, with a reference voltage set at a level that divides the input range into 16 equal segments.

In the decoding stage, the outputs of the op-amps are connected to a combination of logic gates (e.g., AND, OR gates) to generate the 16 output lines. Each combination of input bits corresponds to one output line. When an input bit pattern matches the selected combination, the corresponding output line goes high, while the others remain low.

In the output stage, the 16 output lines are available for external connections and further processing.

Circuit Diagram:



Decoder Truth Table:

A	B	C	D	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Y11	Y12	Y13	Y14	Y15
0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
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1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Steps Of Project:

- Designing Logical NOT Gate Circuit
- Designing Logical AND Gate Circuit
- Designing 4 to 16 Backbone Decoder Circuit
- Connecting Gates In Backbone Circuit With BUBBLES
- Connecting All The 16 AND Gates To A Common Terminal
  And Convert High(10V) into Output Pin Number Using Multiplier

Designing AND Gate Using Op-Amps:

AND Gate uses the principle of logical multiplication while operating. If any of the inputs is low, output will also be low. The output will be high only if all the inputs are high.

Working Principle: Here, V3 has been used as the reference, and its value has been selected in such a way that any low input gives negative saturation voltage as output, and all high inputs give positive saturation voltage as output.

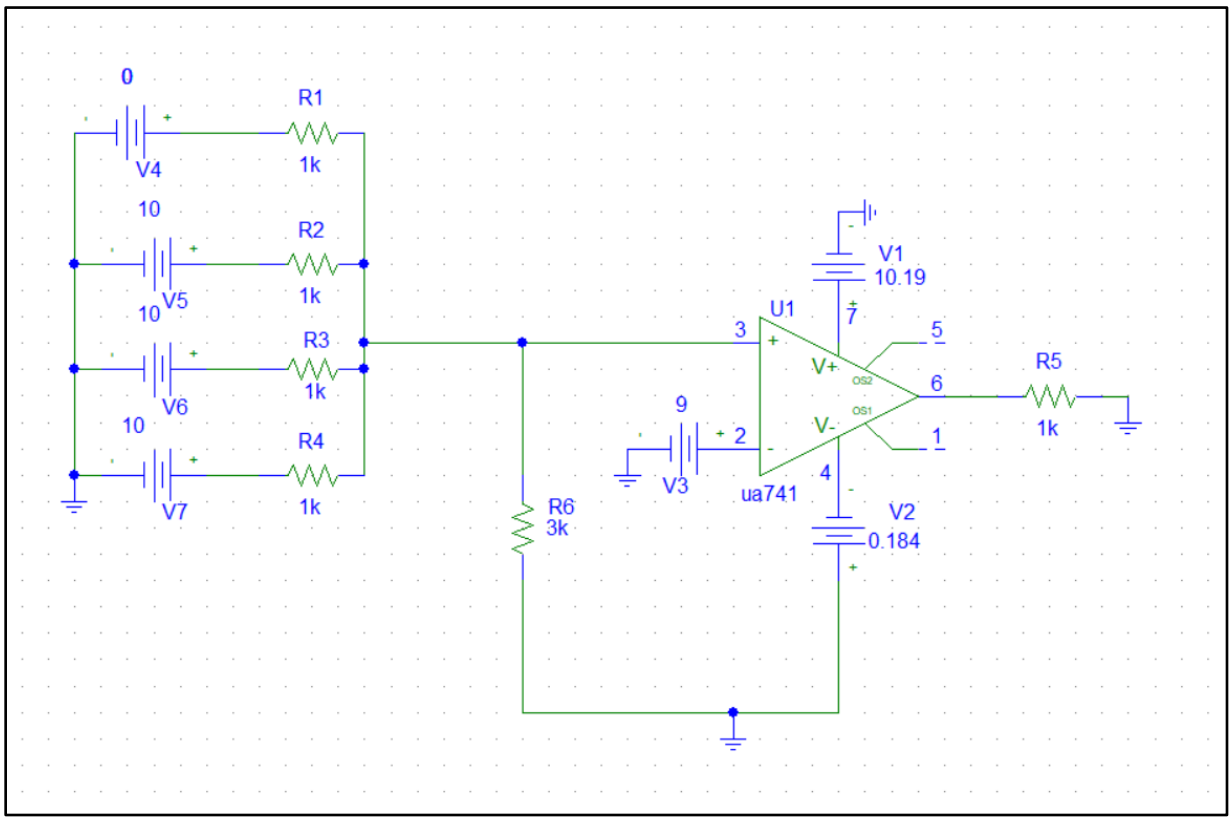
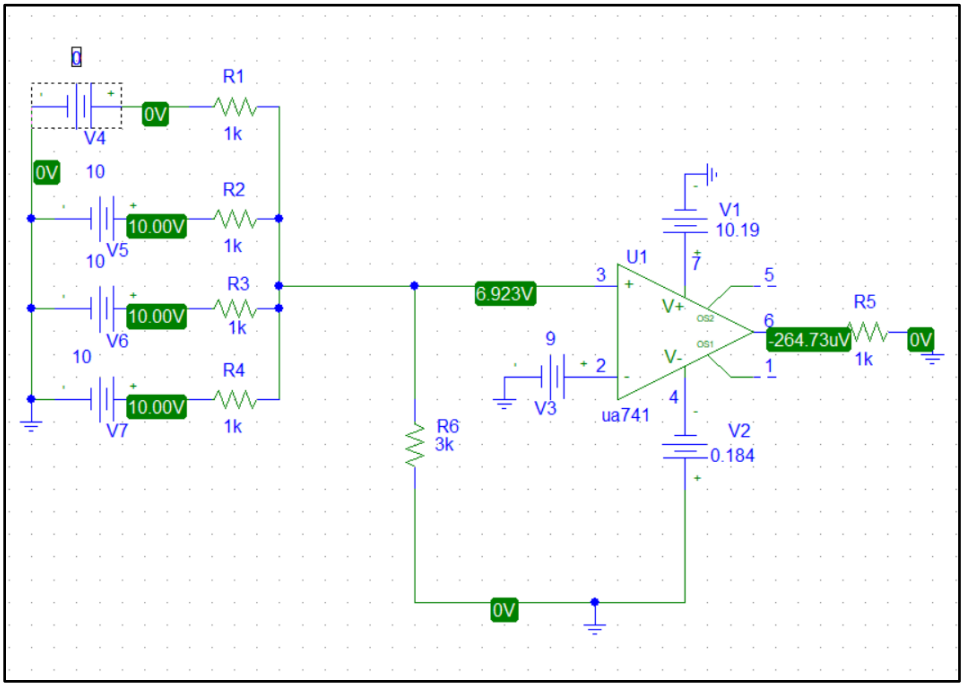
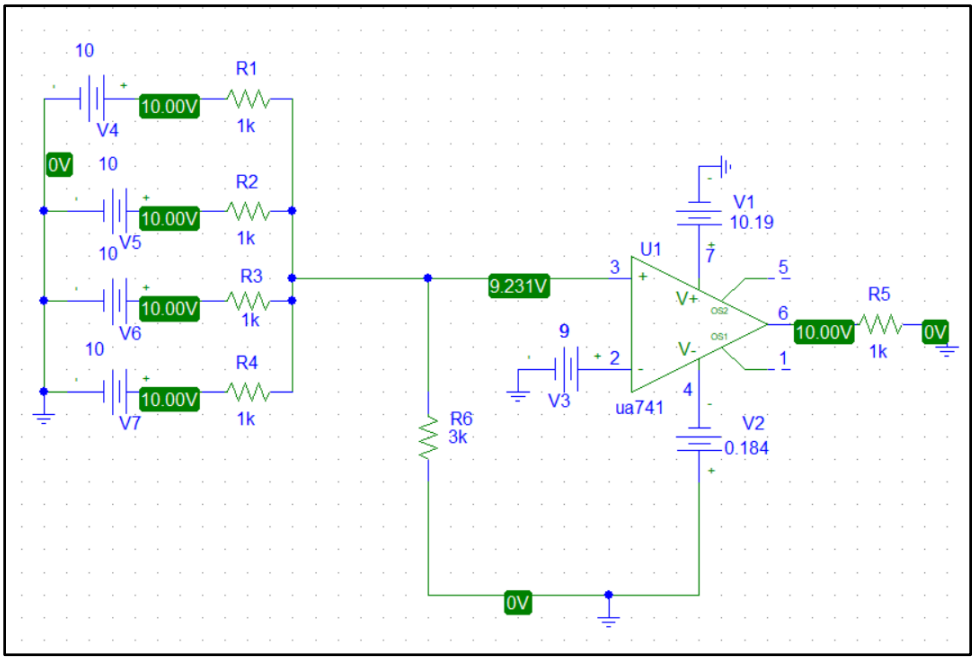


Fig : AND Gate

For one input low, we get low output:



For all high inputs, we get a high output:



Designing NOT Gate Using Op-Amps:

A logical NOT gate logically inverts the input it receives. The output will be high if the input is low and vice versa.

Working Principle: Here, Vref is set in such a way that high (10V) input gives negative saturation voltage as output, and low (less than 10V) gives positive saturation voltage as output.

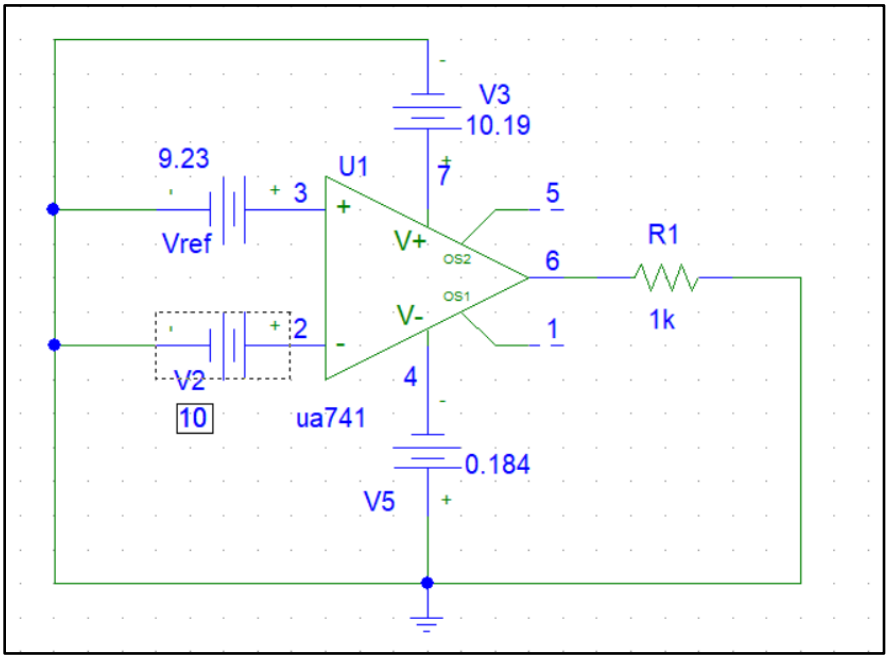
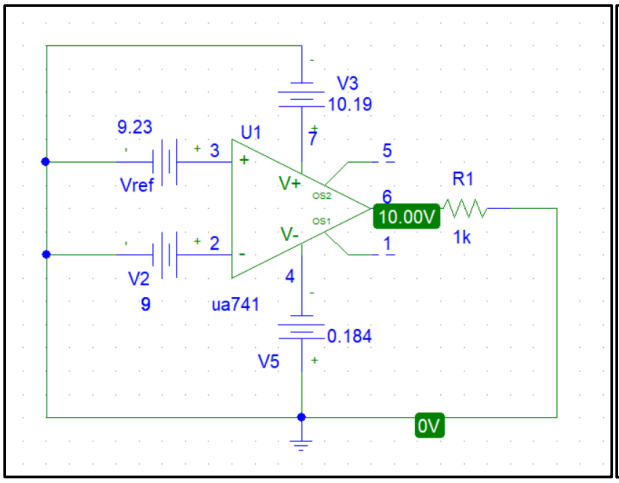
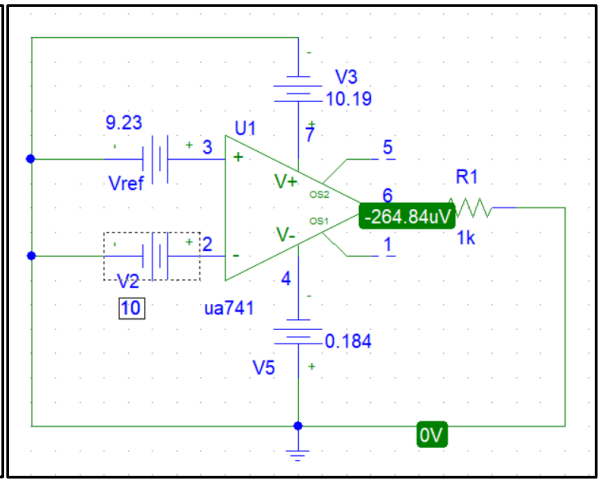
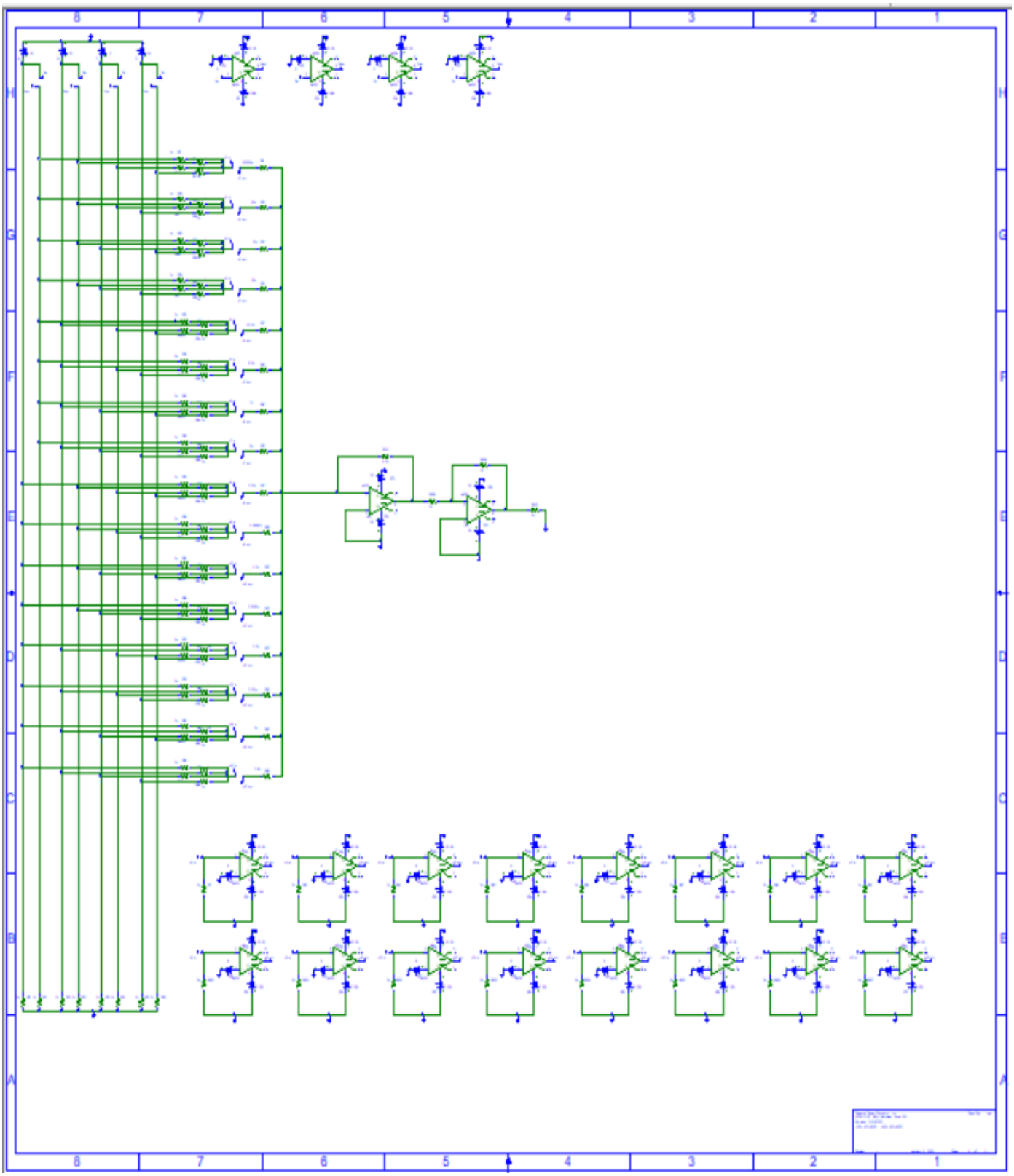


Fig : NOT gate





Main Circuit:



Operating mechanism:

Here, we have made the output voltage to show the terminal which is active for a test case.

The mechanism followed to do so is explained below As a certain terminal will be on 10 V for a single test case the corresponding resistance R_n was selected in such a way so that the following relation is maintained.

$n = 10 * (-Rf/Rn)$

Where,

R_f = Feedback resistance

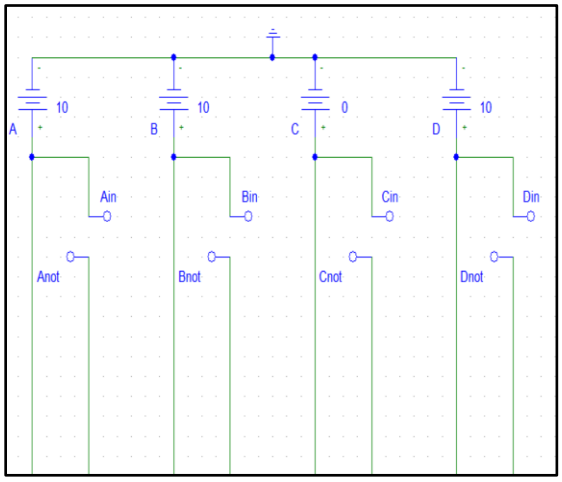
n = number of terminal which is active for a certain test case

As the output will be inverted so we used an inverter circuit further so we get the terminal number as output.

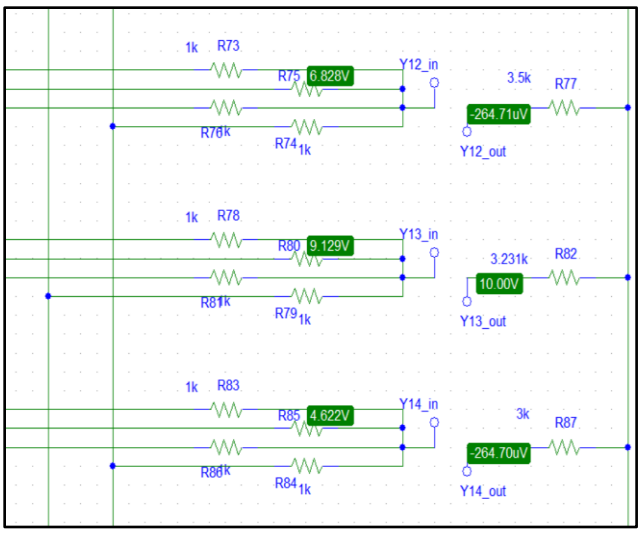
As for example,

For input 1 1 0 1, terminal 13 will be active. So, the output voltage should be 13 .

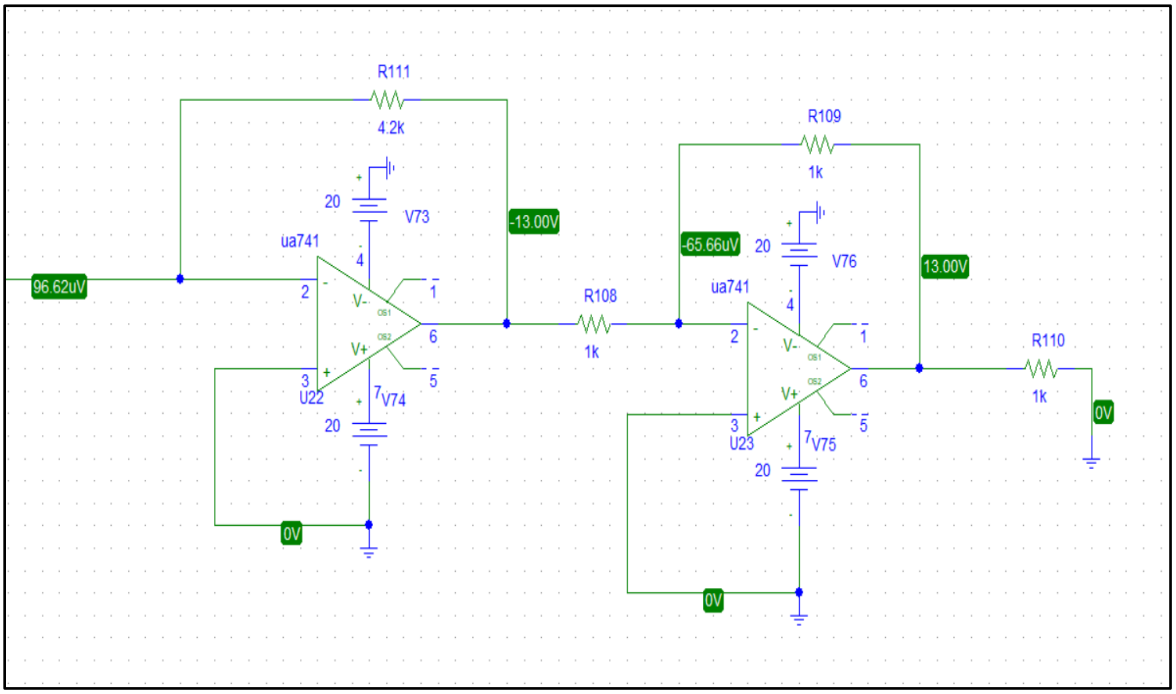
Test Case : 1 1 0 1



Input 1 1 0 1

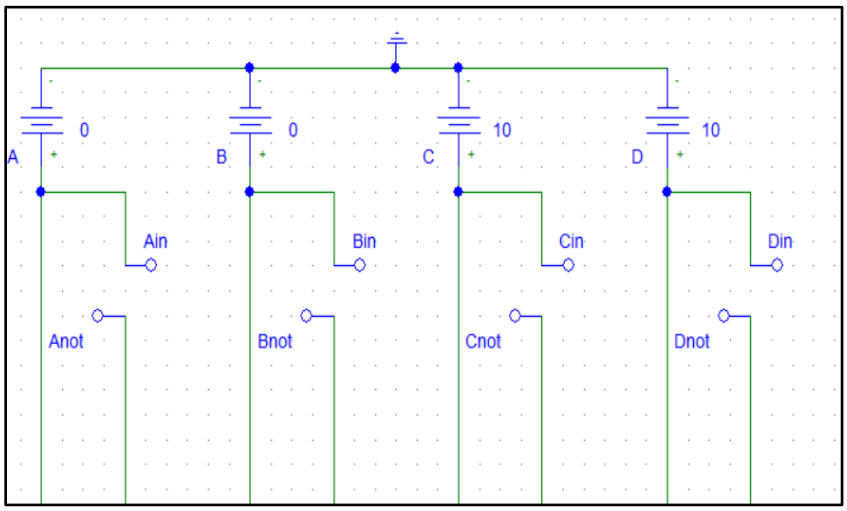


Output High at Pin 13

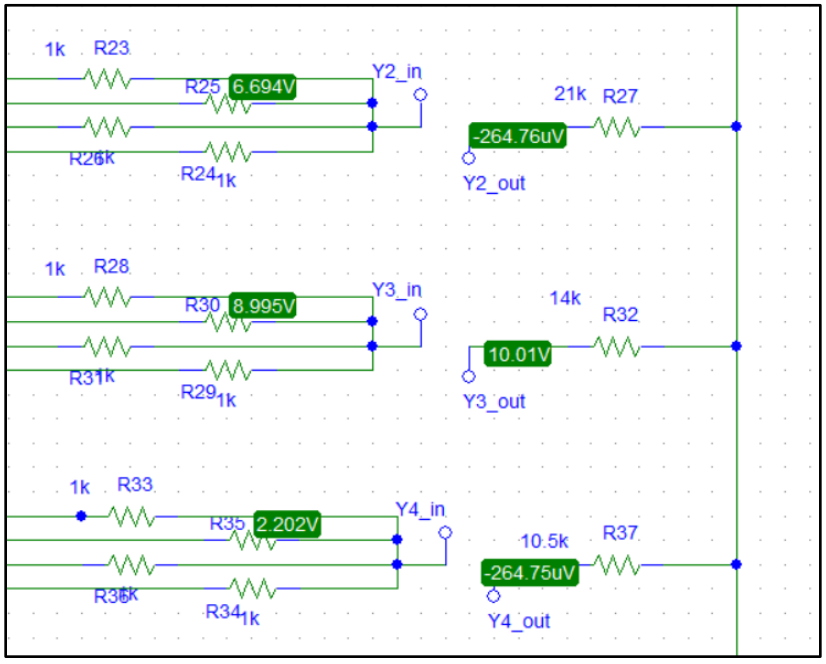


Showing 13 V as the 13th Pin has Logical 1

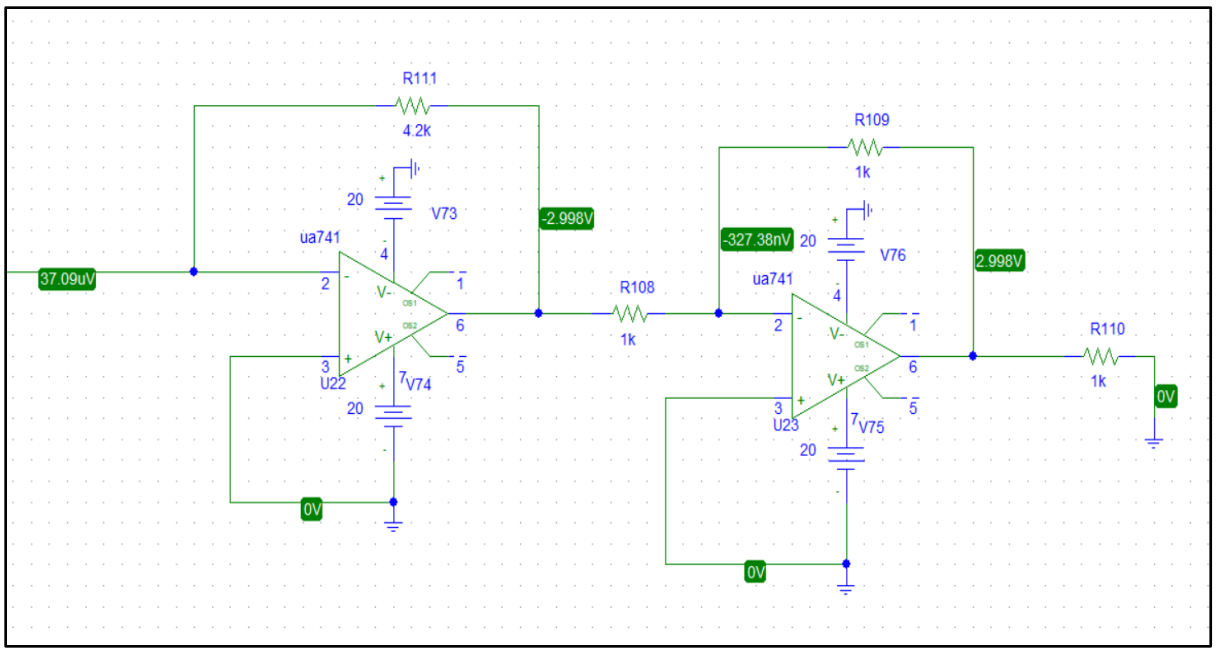
Test Case: 0 0 1 1



Input 0 0 1 1

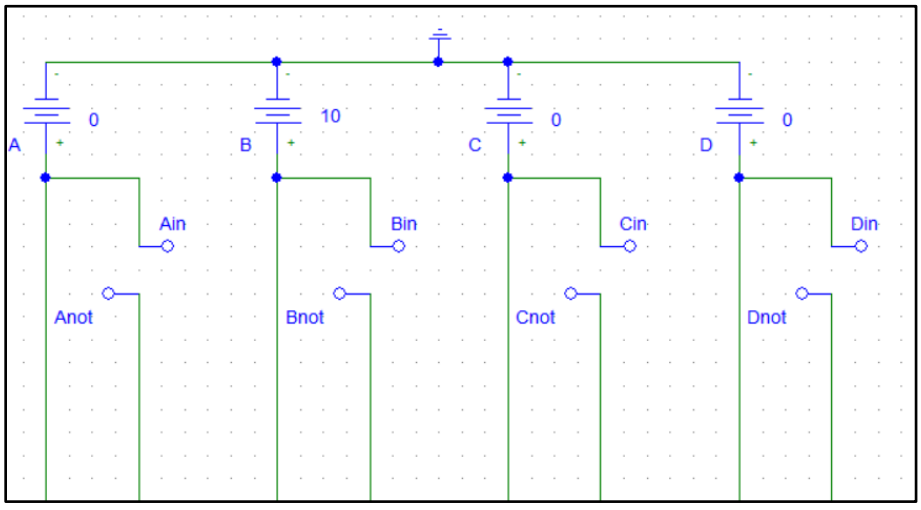


Output High on 3rd Pin

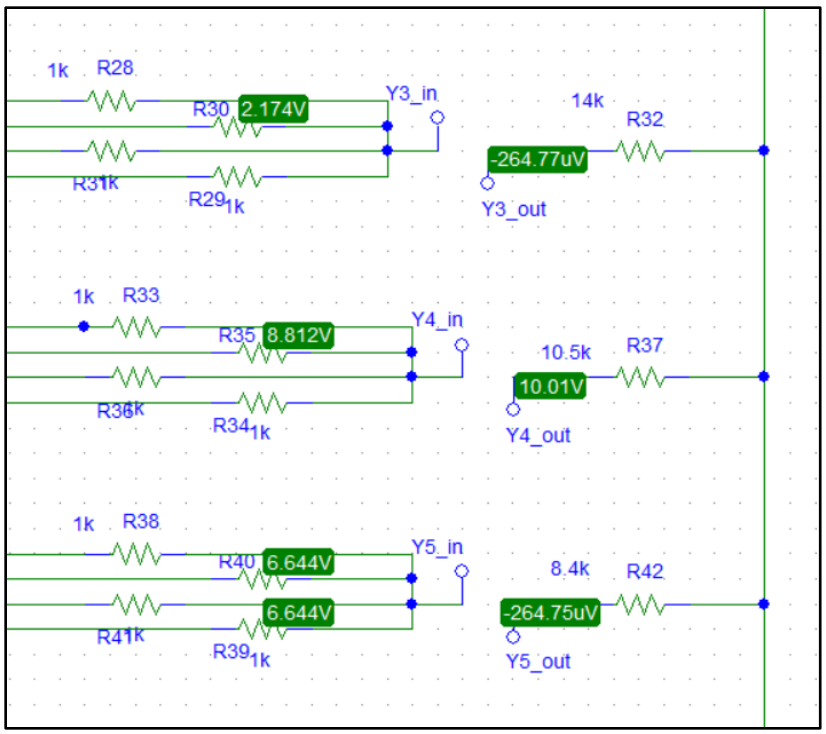


Showing 2.998V as 3rd Pin Gives High Output

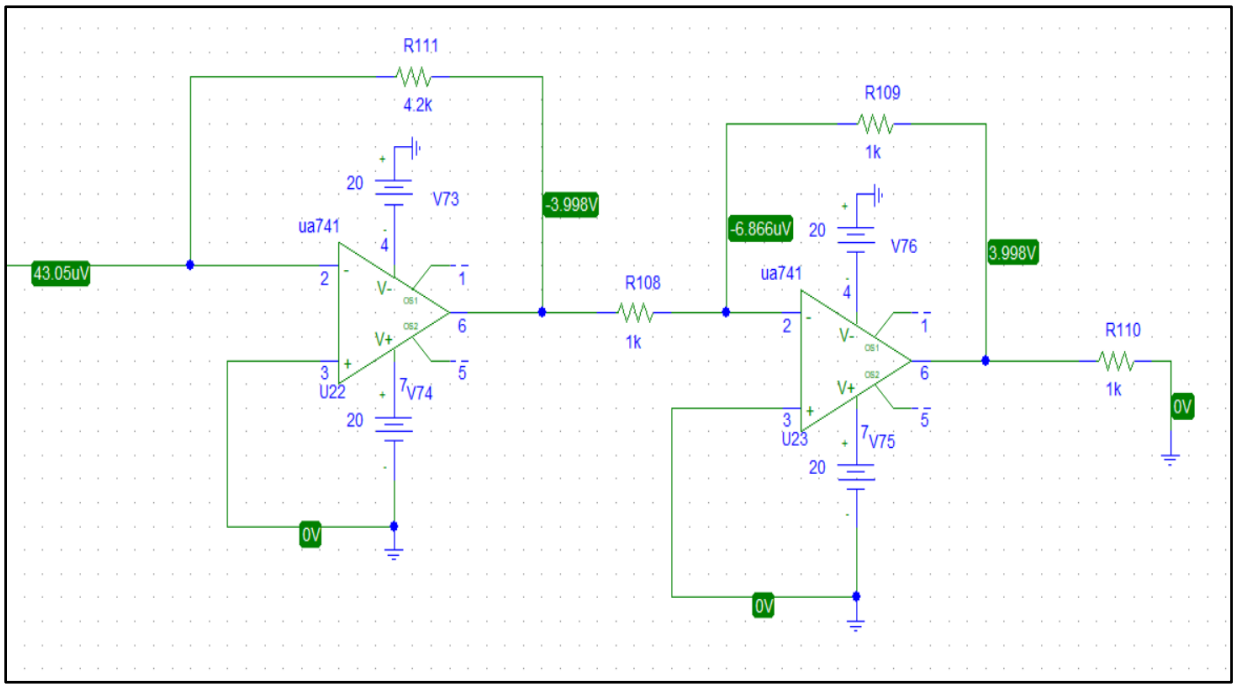
Test Case: 0 1 0 0



Input 0 1 0 0



Output High At 4th Pin



Showing 3.998V as output is high at 4th pin

Application Of The Project:

4 to 16 bit decoders can be applied in the following arenas:

- In computer systems, a 4-to-16 bit decoder can be used to select one of 16 memory locations, allowing the CPU to access different sections of memory, thus used for memory address decoding.

- Decoders are used in LED displays to control which LEDs are turned on or off.

- When combined with a BCD (Binary Coded Decimal) to 7-segment decoder, a 4-to-16 bit decoder can be used to drive a 7-segment display, commonly seen in digital clocks and calculators.

- 4-to-16 bit decoders can be used as building blocks for larger multiplexer (mux) and demultiplexer (demux) circuits, allowing for data routing and selection in digital systems.

- In the field of digital design, programmable logic devices like FPGAs (Field-Programmable Gate Arrays) often use decoders as part of their internal routing and logic configuration.

- In data compression algorithms, decoders are used to decode compressed data back into its original format.

- In remote control systems, like infrared or RF remote controls, decoders are used to interpret the signals received from the remote control unit and translate them into specific actions or commands.

- Decoders can be used in security systems for tasks such as recognizing specific access codes or patterns, allowing or denying access based on the input received.

- In automation and robotics applications, decoders can be used to interpret control commands and select the appropriate actions or movements for robotic systems.

Discussion:

The decoder has been designed assuming 10V as logical 1 and less than 10V as logical 0. The biasing voltage has been set to 10.18V. When it is logical 0 (OFF), the output was supposed to be 0V but rather than perfectly getting 0V, we get around 25µV. Moreover, in some cases of logical 1 (ON) we get 9.96V or 9.98V instead of perfectly 10V. These errors have been caused because of the inability of precise calculation of PSpice. If we could use more precise platforms, the circuit could be expected to work more accurately.